PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

umhackathon@um.edu.my

© UMHackathon 2026

Email: umhackathon@um.edu.my

Table of Contents

1. Project Overview 1
2. Background & Business Objective 1
3. Product Purpose 2
4. System Functionalities 2
5. User Stories & Use Cases 3
6. Features Included (Scope Definition) 3
7. Features Not Included (Scope Control) 3
8. Assumptions & Constraints 4
9. Risks & Questions Throughout Development 4

© UMHackathon 2026

Email: umhackathon@um.edu.my

Project Name: AiyoSafe

Version: 1.0

1. Project Overview

- **Problem Statement:** In Malaysia, many individuals—especially gig workers, students, and low-to-middle-income earners—face severe challenges in making informed financial decisions. Existing budgeting tools are static ledgers; they track what was spent but fail to provide context-aware, forward-looking advice. This reliance on manual tracking or emotional intuition leads to poor cash flow management, missed bills, high-interest debt traps, and vulnerability to financial risks.
- **Target Domain:** AI for Economic Empowerment & Decision Intelligence
- **Proposed Solution Summary:** AiyoSafe is an AI-powered financial decision-support engine. It removes the friction of complex financial planning by acting as a highly personalized, context-aware advisor. Users input their financial snapshot and strict personal constraints (e.g., "Must keep RM500 buffer"). The system then generates a dynamic "AiyoSafe Score" and calculates the quantifiable economic impact of every financial decision, empowering vulnerable Malaysians to proactively protect their wealth and optimize their cash flow.
-

2. Background & Business Objective

- **Background of the Problem:** For years, personal finance has followed a reactive pathway: users spend money, check their bank balance, and attempt to adjust their behavior post-mortem. Professional financial planning is a luxury service inaccessible to everyday citizens. When faced with a high-pressure financial choice (e.g., "Can I afford

© UMHackathon 2026
Email: umhackathon@um.edu.my

to rent this condo?"), users lack the analytical tools to simulate how that decision impacts their specific economic runway.

- **Importance of Solving This Issue:** By shifting from reactive tracking to proactive "Decision Intelligence," we can directly prevent economic loss. AiyoSafe allows everyday users to test financial decisions before committing, avoiding late fees, reducing debt accumulation, and making smart resource allocations without needing financial literacy training.

- **Strategic Fit / Impact:** This perfectly aligns with the domain of Economic Empowerment. By utilizing advanced LLM reasoning to interpret structured data and generate localized, context-aware recommendations, AiyoSafe acts as a digital equalizer, providing enterprise-grade financial intelligence to individual citizens.

3. Product Purpose

3.1. Main Goal of the System: To provide an empathetic, highly localized AI financial decision engine that actively protects a user's cash flow, respects their personal constraints, and quantifies the economic impact of their daily decisions.

3.2. Intended Users (Target Audience):

- University Students & Young Adults managing allowances/loans

4. System Functionalities

4.1. Description: The system operates as a rapid decision-intelligence dashboard. Utilizing a decoupled frontend architecture, the system stores a user's persistent financial context locally and securely stitches it to real-time queries. The AI processes these variables to generate actionable insights, dynamic health scores, and quantifiable savings metrics via a strict JSON output.

4.2. Key Functionalities:

- **Context-Aware Financial Profiling:** Captures income, expenses, and strict user constraints (e.g., "Prefer halal merchants") to serve as the absolute baseline for all AI reasoning.

- **Dynamic AiyoSafe Scoring:** An intelligent, non-static health score (0-100) that adjusts based on recent activities, accompanied by AI-generated "Explainability Microcopy" that explicitly tells the user *why* the score moved.

- **Impact Tracking Dashboard:** Quantifies the LLM's economic value by calculating and displaying estimated RM saved, late fees avoided, and time saved on manual budgeting.

- **"Decide Lah!" Engine:** A query interface where users can simulate financial scenarios and receive mathematically justified approvals or rejections based on their constraints.

4.3. AI Model & Prompt Design

4.3.1. Model Selection: Google Gemini 2.5 Flash. This model was chosen for three critical reasons: 1) Its ultra-low latency inference, essential for real-time dashboard updates; 2) Its superior ability to strictly adhere to complex JSON-schema generation; and 3) Its deep understanding of localized Malaysian context (Manglish, local financial slang), which is vital for user accessibility.

© UMHackathon 2026

4.3.2. **Prompting Strategy:** We utilize a Constraint-Based Master Prompting strategy. The system instruction is heavily engineered to force the AI into a "Financial Optimizer" persona. We explicitly inject the user's [USER FINANCIAL PROFILE] into every prompt, instructing the model to treat those parameters as absolute mathematical boundaries. This prevents generic advice and forces highly specific, localized reasoning.

4.3.3. **Context & Input Handling**: The system maintains context efficiently by utilizing browser localStorage for the financial baseline. Instead of passing massive, token-heavy transaction histories, the system stitches the static profile to a [RECENT ACTIVITY] block. Extremely long user queries are truncated, and the system focuses strictly on extracting numerical data and constraints to prevent context window overflow.

4.3.4. **Fallback & Failure Behavior:** If the LLM generates a hallucinated response, violates the mandatory JSON schema, or times out, the backend parsing logic catches the error. The UI gracefully degrades, displaying a safe fallback state: "AiyoSafe Engine is recalibrating your data. Please try again." Furthermore, prompt guardrails are in place to block the processing of any queries involving self-harm or physical threats.

5. User Stories & Use Cases

- **User Stories:** "As a university student, I want the system to explain exactly *why* my financial score dropped this week, so I know exactly what spending habit to fix."
- **Use Cases (Main Interactions):**

  Progressive Onboarding: User enters income, debt, and constraints via a glassmorphism UI.

  Scenario Simulation: User asks, "Should I buy a RM 200 concert ticket today?" AiyoSafe cross-references the RM500 buffer constraint, does the math, and advises against the purchase to prevent economic instability.

© UMHackathon 2026

Dashboard Generation: The system recalculates the AiyoSafe Score, updates the "Runway Days," and updates the "RM Saved" metrics based on the accepted recommendations.

- **Use Case Diagram:** *(Optional but encouraged for participants to visualize the interaction)*.

6. Features Included (Scope Definition)

- A Progressive Disclosure Financial Snapshot onboarding flow.
- Localized context management via localStorage.
- A dynamic dashboard displaying AiyoSafe Score, Sub-scores, and Runway Days.
- An "Impact So Far" tracker detailing RM saved and fees avoided.
- Context-aware explainability text that links score changes to specific user actions.

7. Features Not Included (Scope Control)

- Direct API integration with live Malaysian banks (Open Banking).
- Complex, long-term wealth management algorithms (e.g., stock market or crypto portfolio optimization).
- Multi-user or family account syncing.

8. Assumptions & Constraints

- **LLM Cost Constraint:** Token costs are kept highly manageable by utilizing Gemini Flash over heavier models and by employing a "Stitched Context" architecture. Instead of

© UMHackathon 2026

sending thousands of rows of transaction data, we condense the user's state into a lightweight summary block on the client side before inference.

- **Technical Constraints:** The current MVP does not pull live bank data; it relies on user-provided inputs or simulated scenarios to demonstrate the AI's decision-making logic.
- **Performance Constraints:** The fluidity of the dashboard rendering is dependent on API latency. Therefore, the UI relies on asynchronous loading states to maintain a high-quality user experience during the 1-3 second LLM calculation window.
- **User Input:** The model is unable to generate without human prompts and is not ready for production so requires human approval.

9. Risks & Questions Throughout Development

- **Mathematical Hallucinations:** How do we ensure the AI does not invent incorrect math when calculating the user's runway?*Mitigation:* Strict system instructions force the AI to break down its reasoning logically before outputting the final JSON integer, greatly reducing mathematical errors.

- **Data Accuracy:** How do we provide value if the user estimates their expenses inaccurately?*Mitigation:* AiyoSafe focuses on the *logic* of the decision based on the constraints provided, empowering the user to learn better decision-making habits regardless of perfect data accuracy.

© UMHackathon 2026